

Self-adaptive logit balancing for deep neural network robustness: Defence and detection of adversarial attacks

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ABSTRACT

With the widespread applications of Deep Neural Networks (DNNs), the safety of DNNs has become a significant issue. The vulnerability of the neural networks against adversarial examples deepens concerns about the safety of DNNs applications. This paper proposed a novel defence method to improve the adversarial robustness of DNN classifiers without using adversarial training. This method introduces two new loss functions. First, a zero-cross-entropy loss is used to punish overconfidence and find the appropriate confidence for different instances. Second, a logit balancing loss is proposed to protect DNNs from non-targeted attacks by regularising incorrect classes' logits distribution. This method achieved competitive adversarial robustness compared to advanced adversarial training methods. Meanwhile, a novel robustness diagram is proposed to analyse, interpret and visualise the robustness of DNN classifiers against adversarial attacks. Furthermore, a Log-Softmax-pattern-based adversarial attack detection method is proposed. This detection method can distinguish clean inputs and multiple adversarial attacks via one multi-classification MLP. In particular, it is state-of-the-art in identifying white-box gradient-based attacks; it achieved at least 95.5% accuracy for classifying four white-box gradient-based attacks with maximum 0.1% false positive ratio.

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1. Introduction

DNNs have achieved impressive results in vast areas, such as computer vision, robotics, and natural language processing. The concerns about the safety of DNNs are rising because the DNNs are potentially applied to safety-critical areas, for example, the perception modules and end-to-end controllers for self-driving cars or autonomous robotics [1,2]. Moreover, the DNNs are significantly vulnerable to adversarial perturbations - the small input changes that can cause mistakes to the networks [3,4]. In general, the inputs modified by adversarial perturbations are also known as adversarial examples [4,5].

The existence of adversarial examples is a challenge to the robustness and trustworthiness of DNNs applications. An excessive number of methods were proposed to attack DNNs by generating adversarial examples from different perspectives. However, designing and developing defence algorithms for DNNs is more challenging than attacking them; furthermore, it is complicated to evaluate a defence method [6].

Adversarial training is one of the most intuitive and widely used methods, which can train DNNs to have higher adversarial robustness by introducing the adversarial examples into training datasets [5]. The major limitation of the adversarial training methods is that they can only defend the specific kinds of adversarial examples used during the training process [5]. Meanwhile, the adversarial training methods are time-consuming and computational power costly. Besides proactively training robust models against adversarial attacks, developing detectors to recognise whether the inputs are adversarial examples is another direction to make DNNs more trustworthy and safer.

To address the security issues of DNNs, we proposed a novel adversarial defence method and an adversarial example detection method that can improve the adversarial robustness of DNNs proactively and reactively. The main contributions of our works are summarised as the followings:

- A novel regularisation method is proposed to significantly improve the general robustness of DNNs against various adversarial attacks. Regularising the Log-Softmax values of incorrect classes by reducing their standard deviation can effectively defend against non-targeted adversarial attacks.

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- Without depending on adversarial training, the proposed defence method needs less computational cost and training time to achieve competitive robustness compared with advanced adversarial training algorithms, such as TRADES [7] and adversarial weight perturbation (AWP) [8].
- A novel robustness diagram is proposed to analyse, interpret and visualise the robustness of DNN classifiers against adversarial examples. The robustness diagrams plot the expected attack success rate of modified inputs as a function of confidence perturbation of adversarial examples.
- A Log-Softmax-pattern-based detection method is proposed to distinguish between clean inputs and various adversarial examples and recognise their attack algorithms of adversarial example generation.

2. Related Work

2.1. Adversarial Perturbation

Adversarial perturbations in the computer vision field are the small changes or modifications to inputs that are often imperceptible to the sense of human sight and will cause the machine learning models to make mistakes [4,5,9]. The input images that are modified by the adversarial perturbations are also known as Adversarial Examples (example illustrated in Fig. 1) [4,5].

According to Szegedy et al. [4], the issue of vulnerability to adversarial examples is impairing the performance of several machine learning models, including the newest DNNs. Moreover, similar adversarial examples can often mislead various classifiers, which are trained under different datasets and have distinct architectures [4,5]. Using adversarial examples against DNNs is a commonly used method for verifying the robustness of DNN applications such as facial landmark detection [10], medical imaging [11], and self-driving vehicles [12].

The fast gradient sign method (FGSM) developed by Goodfellow et al. [5] is a fundamental algorithm that generates adversarial examples by adding linear perturbation to inputs. The adversarial example of input x can be denoted by $x_{adv} = x + \epsilon \times \text{sign}(dL/dx)$, where ϵ is the scale of perturbation, dL/dx is the gradient of the loss function, and sign is the signum operation [5]. When setting $\epsilon = 64/255$, FGSM can make the shallow Softmax classifier output an error rate at 99.9% with an average confidence of 79.3% on the MNIST [13] and cause a maxout network misclassified 89.4% of the adversarial examples with 97.6% average confidence [5]. Moreover, Szegedy et al. [4] proposed a classic method to find adversarial examples around the original inputs using a box-constrained L-BFGS optimiser.

Based on one-step methods like L-BFGS [4] and FGSM [5], it is intuitive to generate adversarial examples by taking multiple small steps iteratively to improve the effectiveness and search coverage.

Basic Iterative Method (BIM) [14] and Projected Gradient Descent (PGD) [15] are designed using the iterative technique. By integrating the momentum term into the iterative attack process, Dong et al. [16] proposed a method that can stabilise update directions and escape from poor local maxima during the iterations, resulting in more transferable adversarial examples. Furthermore, to improve the transferability of adversarial examples, Yuan et al. [17] proposed a novel architecture called Meta Gradient Adversarial Attack (MGAA), which is plug and play and can be integrated with any existing gradient-based attack method.

In addition to white-box attacks represented by the gradient-based attack methods, there are also many black-box attacks, for instance, the Square Attack [18]. Compared to adversarial attack methods, designing and developing defence algorithms for DNNs is more challenging; furthermore, it is also complicated to evaluate and prove the effectiveness of a defence method [6].

2.2. Proactive Defence Method

The methods for defending against adversarial attacks are designed based on two simple strategies. The proactive defensive strategy is to build robustness towards adversarial examples into the DNNs themselves, thus reducing or eliminating the influence of adversarial examples. The reactive strategy is to design detectors that can distinguish clean inputs and adversarial examples and then reject the results of adversarial attacks.

Adversarial examples can attack DNNs for testing and evaluating their robustness and stability, meanwhile can improve the performance of DNNs as a regulariser [5]. Taking the contribution of Goodfellow et al. [5] as the basis, several advanced adversarial training algorithms were developed, such as Ensemble Adversarial Training [19], Robust Optimisation Adversarial Training [20], Virtual Adversarial Training [21] and TRADES [7]. However, the adversarial training methods share two common drawbacks: efficiency and generalisation problems [22,23].

It takes 3–30 times longer to learn robust networks with adversarial training methods than to learn non-robust plain networks [22]. The limitation of training efficiency makes adversarial training methods challenging in applications where online learning is necessary. Meanwhile, the trade-off between robustness and generalisation is inevitable for adversarial training methods [23]. Adversarially trained models to have the robustness to specific attacks, e.g., l_∞ adversarial examples, can be misled easily by different types of attacks with other l_p norms. Furthermore, combining perturbations with different l_p norms in adversarial training can not solve this problem [24].

Logit paring is a regularisation method for training robust DNNs, which can be used with or without adversarial training [25]. The core idea of logit paring is to encourage the logits from two paired images to be similar and receive a penalty for overcon-

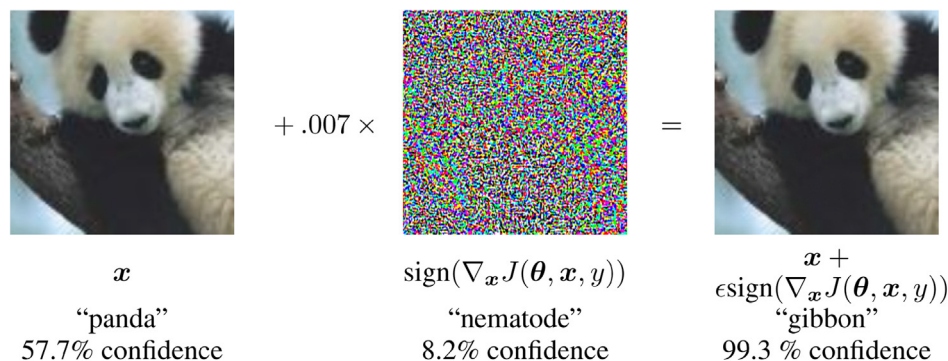


Fig. 1. A classical example of Adversarial Perturbation [5].

fidet [25]. The adversarial logit pairing matches the logits from a clean input x and its corresponding adversarial example x' . In clean logit pairing, x and x' are randomly selected clean training samples, which do not have to be from the same class. However, the clean logit pairing mechanism does not care whether the randomly paired training examples are appropriate and fragile under well-designed attacks [26].

Label smoothing was initially proposed by Szegedy et al. [27] for improving the performance of DNNs, which use smoothed uniform label vectors in place of one-hot label vectors. Let y_{hot} denote the one-hot label vectors of training data with K classes. The smoothed uniform label vectors y_{LS} can be calculated using the equation $y_{LS} = (1 - \epsilon) \times y_{hot} + \epsilon/K$, where ϵ is the hyper-parameter that determines the level of smoothing. It has been proved that a model trained with label smoothing can invalidate some simple gradient-based attacks [27–29]. However, signing uniform label vectors to varying data instances is inappropriate for general robustness and fails under solid attacks [30].

2.3. Adversarial Example Detection

Metzen et al. [31] proposed improving the targeted network's adversarial robustness with a sub-network, which is a binary classifier for detecting adversarial examples and clean inputs. They showed that adding such a classifier to the internal layers of the targeted network and using adversarial training can detect adversarial examples generated using FGSM and BIM methods [31]. However, Lu et al. [32] later showed that this approach is still vulnerable to adaptively designed counterattacks.

Lu et al. [32] assumed that adversarial examples produce different patterns of ReLU activations in the late stages of neuron networks from what is produced by clean images. Based on this hypothesis, they proposed to attach a Radial Basis Function SVM classifier to the targeted models. The SVM takes discrete codes computed by the late ReLU layers of the network as inputs for detecting perturbations in the test image [32]. This detection method can effectively distinguish adversarial examples generated by FGSM and BIM from clean inputs [32].

Xu et al. [33] proposed using feature squeezing to detect adversarial perturbations to a clean image. They attached two external models to the targeted network. These models will reduce the colour bit depth of each pixel in the input image and perform spatial smoothing over the image. After that, the predictions of the targeted network over the original image and the feature-squeezed images are compared [33]. If a significant difference is found between the predictions, the image is considered an adversarial example.

Roth et al. [34] proposed an empirical adversarial example detection method based on a statistical test that measures how logits change under adversarial perturbations. Besides recognising adversarial examples, this method can also estimate the true label (class) of the modified inputs [34]. This method is designed based on the phenomenon that standard methods for creating adversarial examples result in perturbations that overfit to specifics of the inputs and are not robust [34]. For instance, when adding noise to the adversarial examples, the classification probability of the true class will be increased.

Hosseini et al. [35] proposed a generic binary classifier that takes logits of clean and perturbed images as input and detects whether the images are adversarial examples. They claim that the binary classifier can identify anomalies in adversarial logits beyond the directionality toward the true label and improve the detection performance. Besides this binary-classifier-based detection method, Hosseini et al. [35] also proposed an attack method called logit mimicry attack (LMA) to bypass the statistical test detector described in [34] and their new binary classifier detector to prove

there is no robust logit-based adversarial example detection method.

3. Proposed Proactive Defence Method

Some parts of this section is in proceedings of the 21st International Conference on Image Analysis and Processing [36].

3.1. Preliminary

In this paper, the adversarial defence problem is simplified to training a DNN as an image classifier to have high test accuracy on clean data and adversarial examples. An N -layers DNN is a mapping $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ that takes $x \in \mathbb{R}^n$ as an input and produces $y \in \mathbb{R}^m$ as an output. It can be expressed as:

$$f(x) = F^N \left(F^{N-1} \dots \left(F^1(x) \right) \right) = y \quad (1)$$

When using a DNN to solve a K -class classification problem, let S denote the output of the Softmax layer, such that:

$$S = \text{Softmax}(f(x)) = \{s_1, s_2, \dots, s_K\} \quad (2)$$

where $0 \leq s_i \leq 1$ and $s_1 + s_2 + \dots + s_K = 1$. The DNN will assign the label of input x according to:

$$\text{predict} = \text{argmax}_i S(f(x)) \quad (3)$$

Most of the DNNs classifiers are trained with the Cross-Entropy (CE) Loss:

$$L_{CE} = L_{CE}(t, S(f(x))) = -\log(s_t) \quad (4)$$

where t is the ground truth label of input x , $t \in K$. It is necessary to have a threat model for evaluating the performance of defence methods. Without a threat model, defence proposals are likely not to be falsifiable or trivially falsifiable [6]. For any clean input x and a similarity metric $\mathcal{D}$, x' is a valid adversarial example if x' is misclassified and $\mathcal{D}(x, x') \leq \epsilon$, where ϵ is perturbation limit. The white-box adversarial attacks can access complete knowledge of deep neural networks and the inner workings of the defence. Meanwhile, black-box attacks, transfer attacks and adaptive attacks are also deployed to evaluate the proposed defence method. The proposed defence method should be robust against non-targeted attacks at different L_p norms.

3.2. Rethink of Vulnerability

Problems of Training Data. One-hot labelling [37] is a standard method to organise the datasets for machine learning because it is efficient and straightforward. Under one-hot labelling, every instance of a dataset is labelled as one group of bits, among which only the ground truth is encoded as 1, the single high, and all others as 0. However, the one-hot label may create noises and biases because there is insufficient information within every data instance to support that instance is 100% true to be in the labelled class. Under ideal conditions, every instance in the dataset should be labelled individually according to its feature quality. However, labelling the data so finely is almost impossible.

Problems of Learning Rules. It is intuitive to utilise 0–1 loss when training a machine learning classifier with a one-hot labelled dataset. However, the problem of adversarial defence is challenging because the 0–1 loss is NP-hard to optimise, and the surrogate loss has too little theoretical guarantee on the tightness [7]. Meanwhile, the current learning rules, such as backpropagation with gradient descent, force the DNNs to give all target labels 100% confidence. Therefore, the DNNs are easily overconfident under the fixed training dataset and under-fitted to solve the general problem [4,5]. The DNNs cannot capture all meaningful features from the limited

database and assign correct weights to every feature. This issue is hard to be resolved by a more profound or more complex network structure [5].

3.3. Self-adaptive Logit Balancing

3.3.1. Zero-Cross-Entropy Loss

In the proposed adversarial defence method, a zero-cross-entropy loss is introduced to solve the overconfidence issue by making the loss equal to 0 when the prediction is equal to its target label. Instead of forcing DNNs to have 100% confidence in every training example towards its target label, the zero-cross-entropy loss only encourages DNNs to believe that the training examples are most likely to be in their labelled classes. In other words, give the largest Softmax value to the target label. The proposed zero-cross-entropy loss can be expressed as follow:

$$Loss(x) = \begin{cases} L_{CE}(t, S(f(x))) \times 0 & \text{if } \text{predict} = t \\ L_{CE}(t, S(f(x))) & \text{if } \text{predict} \neq t \end{cases} \quad (5)$$

Meanwhile, the zero-cross-entropy loss introduces a self-adaptive early stop mechanism to punish the overconfidence, which will help the DNNs to learn more appropriate weights for all extracted features. During training progress, the instances with more apparent features toward the true label will be more likely to achieve 0 loss (correct classification) at the earlier training stage. The DNNs are trained with mini-batches; the loss of clear instances will keep decreasing until all examples in their mini-batch reach 0 loss. Therefore, with the 0-cross-entropy loss, the instances with more features of their corresponding classes will have lower losses than the poor-quality examples.

3.3.2. Logit Balancing Loss

The Log-Softmax values of incorrect labels are not regularised during the standard training process when the costs are calculated only relying on the target label. Most non-targeted white-box attacks try to decrease the target label's confidence by modifying the inputs to have smaller Log-Softmax values corresponding to their true labels. When the confidence of the target label has decreased, the sum of the confidence of other labels will increase because the sum of all confidences has to be 1 when using the Softmax activation function. Suppose the additional confidence reallocated from the target label to incorrect labels can be distributed evenly. In that case, the model is hard to attack until the confidence of the correct label drops to $1/K$ (K is the number of classes).

Therefore, squeezing the distribution of Log-Softmax values of all incorrect classes to their mean could increase the adversarial robustness. Then a novel logit balancing loss [36]:

$$Loss_{LB} = \beta \times SD((-log(S(f(x_i))) \setminus s_t)) \quad (6)$$

is introduced to control the distribution of the Log-Softmax of all incorrect labels, where β is the logit balancing weight and SD stands for standard deviation. Decreasing the standard deviation of the incorrect labels' Log-Softmax values has two effects. The first is making the distribution of the incorrect labels' logits closer to their mean, thus improving the adversarial robustness of DNNs. The second effect decreases the mean of the Log-Softmax values of all classes, increasing the target label's loss value. Therefore, heavy and fixed logit balancing weight will lead DNNs to give insufficient confidence to correct labels. Finally, the logit balancing will lead the model to provide the same confidence for every class; for example, a ten-class data will have 0.1 Softmax confidence for all classes. Therefore, a self-adaptive regularisation parameter is introduced [36]:

$$Loss_{LB} = \beta \times SD((-log(S(f(x_i))) \setminus s_t)) \times s_t \quad (7)$$

where s_t is the Softmax value of the target label. So, the logit balancing loss of each training example will be changed based on its confidence in the target label - the higher confidence of the correct label, the less attenuation of logit balancing loss.

3.4. Gaussian Noise Boost Training With Self-adaptive Logit Balancing

Data randomisation methods such as random resizing and adding random padding can help to improve the adversarial robustness of DNNs [38,39]. Gaussian noise boost training [40] is utilised to solve the limited training dataset's data distribution problem [41]. Instead of training the DNNs on the fixed clean images, Gaussian noise boost training will add different random Gaussian noise to each training example for every training epoch.

Algorithm 1: Self-Adaptive Logit Balancing Training [36]

Data: Logit balancing weight β , batch size m , network architecture parametrised by θ , Randomly initialise network f_θ , Gaussian noise weight σ .

Result: Robust network f_θ .

```

begin
  Read mini-batch  $B = (x_1, x_2, \dots, x_m)$  from training set, and its
  corresponding target label  $T = (t_1, t_2, \dots, t_m)$ 
  repeat
    for  $i = 1$  to  $m$  do
       $GN_{x_i} = x_i + \sigma \times \text{GaussianNoise}$ 
      if  $\text{argmax}_{index} S(f(GN_{x_i})) = t_i$  then
         $loss_i = \beta \times SD((-log(S(f(GN_{x_i}))) \setminus s_t)) \times s_t$ 
      else
         $loss_i = L_{CE}(f_\theta(GN_{x_i}), t_i) = -log(s_t)$ 
      Update:  $\theta \leftarrow \theta - \sum_{i=1}^m \nabla[loss_i] \div m$ 
    until training is converged.
```

As described in Algorithm 1, the proposed adversarial defence algorithm is designed by integrating Gaussian noise boost training and self-adaptive logit balancing. This defence algorithm suggests minimising Eq. 7 to maximise the robust accuracy under adversarial attacks. The self-adaptive logit balancing mechanism is activated if the DNN can classify the current input correctly, thereby avoiding the negative effect of logit balancing and maximising the overall accuracy.

3.5. Experiments and Results Evaluation

3.5.1. Experiments Setup

Logit Balancing Model Setting On the MNIST data, the CNN architecture in TRADES [7] is used for a fair comparison, which has four convolutional layers, followed by three fully-connected layers. We set logit balancing weights $\beta = 0.16$, Gaussian noise weight $\sigma = 0.4$. On the CIFAR-10 data, the model is trained with WideResNet-34-10 architecture [42], which is the same as in TRADE and AWP. We set logit balancing weight $\beta = 0.02$ and Gaussian noise weight $\sigma = 24/255$. The optimiser is Adam, with a learning rate of 0.02 and training for 100 epochs.

Adversarial Training Methods for Comparison Two defence methods with adversarial training are investigated for a detailed comparison with the proposed defence method because they share the same model architecture and use the same benchmark datasets. TRADES is one of the advanced adversarial training algorithms, the first-place winner of the NeurIPS 2018 Adversarial Vision Challenge (Robust Model Track) [7]. AWP is a regularisation method that regularises the weight loss landscape of adversarial training and forms a double-perturbation mechanism that injects

the worst-case input and weight perturbations [8]. The TRADES model on MNIST data is adversarially trained with perturbation $\epsilon = 0.3$. The TRADES and AWP models on CIFAR-10 data are adversarially trained with perturbation $\epsilon = 8/255$.

Attack Algorithms for Robustness Evaluation The models are tested by various attacks under L_∞ and L_2 norms. In this paper, Torchattacks [43] is used to perform all the tests. The Torchattacks is a PyTorch library containing the most popular adversarial attacks to generate adversarial examples for verifying the robustness of deep learning models [43]. The PGD and CW are proposed by Madry Lab MIT [15]. The APGD and APGDT are designed by combining Expectation Over Transformation, and PGD [44,45]. The fast adaptive boundary attack (FAB) has an intuitive geometric meaning, yields quickly high-quality results and is robust to the phenomenon of gradient masking [46]. The square attack is a query-efficient black-box adversarial attack via random search to find adversarial examples [18]. Table 1 provides the default settings of all attack methods.

3.6. Learning Curve

The proposed method has a significant advantage in efficiency over adversarial training methods, for instance, TRADES. On a dual Nvidia GeForce RTX 2080 TI GPU cluster, the proposed method and plain WideResNet take around 90 s for each training epoch. However, the TRADES needs around 1100 s for a training epoch.

Fig. 2 illustrates the learning curves of the logit balancing model and a plain WideResNet model on the CIFAR-10 dataset. As shown in Fig. 2 (a), the standard deviation of incorrect labels' Log-Softmax values of the two models showed opposite results as training proceeded. The Logit-balancing model compresses this value from start to finish. However, this value for the plain WideResNet keeps

Table 1
Attack approaches and their parameters settings.

Attack Method	Attack Parameters
PGD- L_∞	eps = 8/255, alpha = 2/255, steps = 20, random start = True
APGD- L_∞	eps = 8/255, alpha = 2/255, steps = 40, sampling = 10
APGDT- L_∞	eps = 8/255, steps = 100, restarts = 1, eot_iter = 1, rho = 0.75
Square- L_∞	eps = 8/255, size of squares = 0.8, steps = 5000, restart = 1
FAB- L_∞	eps = 8/255, alpha = 0.1, steps = 100, restart = 1
PGD- L_2	eps = 1.0, alpha = 0.2, steps = 40, random start = True
CW- L_2	c = 1.0, kappa = 0, steps = 1000, lr = 0.01

rising with training, even if there is no longer a significant improvement in classification accuracy.

As shown in Fig. 2 (b), during the first 20 epochs, the Logit-balancing model's natural accuracy increases from 40% to nearly 80%, and its robust accuracy under PGD²⁰ attacks remains at 0%. From the 21st epoch, the robust accuracy starts rising, twists and turns to its highest 74.67%, and the natural accuracy rises smoothly and slowly to its highest 85.05% at the 88th epoch. On the other hand, the natural accuracy of plain WideResNet is always a little higher than the Logit-balancing model, but the gap gradually decreases with training. However, the robust accuracy of plain WideResNet under PGD²⁰ attacks stays at 0% through the training process.

3.7. Robustness Evaluation and Comparison

After exploring the logit balancing models with various logit balancing weights and Gaussian noise weights, it is found that increasing the σ of Gaussian noise can improve the overall robustness. However, heavy Gaussian noise will cause the natural accuracy to be impaired. For example, the natural accuracy of the logit balancing model on CIFAR-10 data will drop to 74% from 85% when rising $\sigma = 24/255$ to $\sigma = 64/255$. Meanwhile, larger logit balancing weights can only give the deep learning models gradient masking without actual robustness. Large logit balancing weights will give higher robust accuracy under non-targeted white-box attacks, for example, PGD, but lower robust accuracy under other attacks.

As shown in Table 2, on MNIST data, the logit balancing (LB) model has the overall adversarial robustness that is close to the performance of TRADES when setting perturbation limit $\epsilon = 0.1$. However, when increasing the perturbation limit to $\epsilon = 0.3$, the overall robustness of the logit balancing model is challenged. On CIFAR-10 data (see Table 3), the proposed defence method is more comprehensive for defending both L_∞ and L_2 distance attacks under non-targeted settings, as the adversarial training methods can only defend the attacks used during their training process.

Furthermore, it is found that adding Gaussian noise masking to test inputs can eliminate the effects of adversarial perturbations and improve robustness further. By adding Gaussian noise masking to adversarial inputs, the adversarial robustness of the logit balancing model has been improved significantly against all tested attack algorithms. The Gaussian noise masked inputs results are shown in Table 4.

By adding Gaussian noise masking ($\sigma = 16/255$), the robust accuracy of APGDT increased from 3.97% to 32.64%. The robust

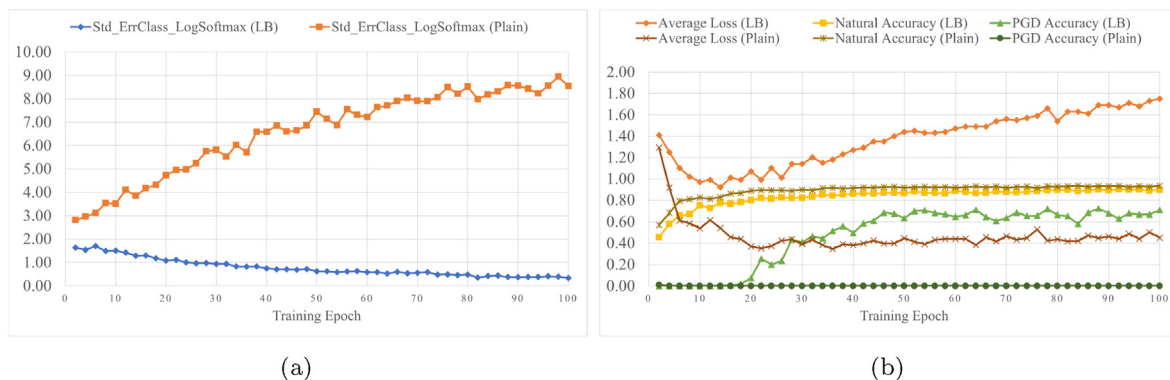


Fig. 2. Learning curve comparison between the Logit balancing model and the plain WideResNet on the CIFAR-10 dataset.

Table 2

Robust accuracy on MNIST data under various attacks [36].

Defence Model	Perturbation	PGD	APGD	Square	FAB	PGD- L_2	CW- L_2
Logit Balancing	$\epsilon = 0.1$	96.81	94.01	94.48	94.58	95.60	96.72
	$\epsilon = 0.3$	56.41	0.6	0.5	18.2	31.88	89.68
TRADES	$\epsilon = 0.1$	98.88	98.82	98.81	98.86	97.19	97.24
	$\epsilon = 0.3$	96.33	9.356	92.93	94.93	82.02	96.51

Table 3

Robust accuracy on CIFAR-10 data under various attacks.

Defence Model	ϵ	PGD	APGD	APGDT	SQUARE	FAB	PGD- L_2	CW- L_2
Logit Balancing	8/255	74.67	48.18	3.97	40.06	53.80	73.30	74.48
	16/255	66.99	6.29	0.0	14.74	38.94	61.02	69.56
TRADES ($1/\lambda = 6$)	8/255	56.61	54.81	55.07	58.91	54.87	28.49	11.28
	16/255	21.08	18.99	17.29	25.55	20.15	3.79	2.04
AWP-TRADES	8/255	60.13	58.58	56.20	60.40	56.84	33.98	22.19
	16/255	27.69	25.58	25.70	30.18	26.02	6.09	5.63

Table 4

Comparisons of logit balancing with adversarial training methods under various attacks with Gaussian masks.

Defence Model	Natural Accuracy	PGD	APGD	APGDT	SQUARE	FAB	PGD- L_2	CW- L_2
Logit Balancing	85.05	77.82	61.58	32.64	73.47	80.25	79.06	83.53
TRADES ($1/\lambda = 6$)	74.53	53.34	61.61	61.22	67.07	67.50	52.49	64.10
AWP-TRADES	76.95	56.91	64.05	63.44	68.68	68.54	50.80	67.57

accuracy of AGPD reached 61.58%, which is almost equal to the accuracy of TRADES (61.61%). Meanwhile, the robust accuracy of Square and FAB attacks increased from 40.06% and 53.80% to 73.47% and 80.25%, which outperformed the TRADES and AWP's accuracy at around 68%. Furthermore, the proposed defence method has even more advantages than TRADES and AWP for defending L_2 norm attacks with or without Gaussian noise masking. Gaussian noise-masking can also slightly improve the robustness of the TRADES and AWP model against adversarial attacks. However, Gaussian noise masking will impede the natural accuracy of TRADES and AWP models by about 10%.

3.8. Results Under Transfer Attacks

Adversarial examples can transfer between models, which means the adversarial examples generated against one model with a specific architecture will be more likely to be adversarial against other models [5]. Adversarial transfer attacks are the adversarial examples generated from source models to attack the target model. Furthermore, such transfer attacks successfully circumvent defences that utilise gradient masking [6].

A comprehensive transfer attacks test was implemented to prove that the adversarial robustness of self-adaptive logit balancing is not from gradient masking. As shown in Table 5, the logit balancing model is attacked by three source models. The natural model is trained with clean data and without defences. The transfer attacks from the natural model can not affect the targeted model. These results prove that the robustness of the logit balancing model does not come from gradient masking. The targeted model can ensure at least 60% robust accuracy under the transfer attacks from TRADES and AWP models.

Table 5

Robustness under transfer attacks [36].

Target Model	Source Model	PGD	APGD	APGDT	SQUARE	FAB	PGD- L_2	CW- L_2
Logit Balancing	Natural Model	79.31	80.66	83.14	84.09	84.73	80.45	85.15
	TRADES Model	59.60	72.03	74.42	82.63	80.08	65.39	78.86
	AWP-TRADES	61.64	74.11	74.03	82.13	79.83	65.77	79.43

3.9. Adaptive Attack and Attack Convergence

Testing the defence via various non-adaptive attacks (without countering the proposed defence mechanism) is necessary but not sufficient [6]. Therefore, it is essential to apply adaptive adversaries designed to counter the proposed defence's details and try to invalidate the robustness claims. The adaptive attack method for logit balancing is proposed by replacing the loss function of regular gradient-based attacks with logit balancing loss - Eq. 6. Then adjust attack parameters until the attack is converged to explore the limitations of the proposed defence method. As illustrated in Table 6, the adaptive attacks can effectively mislead both LB and TRADES. The TRADES is becoming more vulnerable than LB when increasing the attack perturbation and attack iteration.

Table 6

Adaptive attacks and attack convergence.

Defence Model	Perturbation	Attack Iterations	Adaptive PGD- L_∞	Adaptive PGD- L_2
Logit Balancing	$L_\infty = 8/255$	10	58.04	56.59
	$L_2 = 1.0$	100	53.89	54.34
	$L_\infty = 16/255$	10	49.72	47.42
	$L_2 = 2.0$	100	43.21	44.58
	$L_\infty = 32/255$	10	45.07	47.42
	$L_2 = 3.0$	100	32.96	38.99
TRADES	$L_\infty = 8/255$	10	75.31	67.54
	$L_2 = 1.0$	100	75.22	62.07
	$L_\infty = 16/255$	10	59.80	48.66
	$L_2 = 2.0$	100	57.14	34.88
	$L_\infty = 32/255$	10	36.07	35.12
	$L_2 = 3.0$	100	22.89	20.29

4. Robustness Analysis and Visualisation

4.1. Reliability Diagrams

Guo et al. [47] proposed that modern DNNs are often not calibrated; therefore, interpreting the DNN classifiers' predicted confidences as probabilities is often incorrect. Guo et al. [47] also suggested using reliability diagrams and confidence histograms to explain and visualise DNN models' calibration error and prediction reliability. Reliability diagrams (see Fig. 3) are a visual representation of model calibration [47–49], which plot the expected accuracy of input as a function of model predicted confidence.

Fig. 3 illustrates the reliability diagrams of three defence models and one plain model taking clean images of CIFAR-10 as inputs. The model's predictions are divided into ten bins based on the highest confidence score of every testing sample. Then calculate the average confidence and average accuracy of each bin. The confidence histogram at the bottom shows how many test examples are in each bin. The reliability diagrams show that the four tested models are not calibrated. The plain model is overconfident. Its average accuracy is lower than its predicted confidence. In contrast, all defence models have higher accuracy than predicted confidence. Fig. 3 shows that the vulnerability of DNN models against adversarial examples comes from overconfidence.

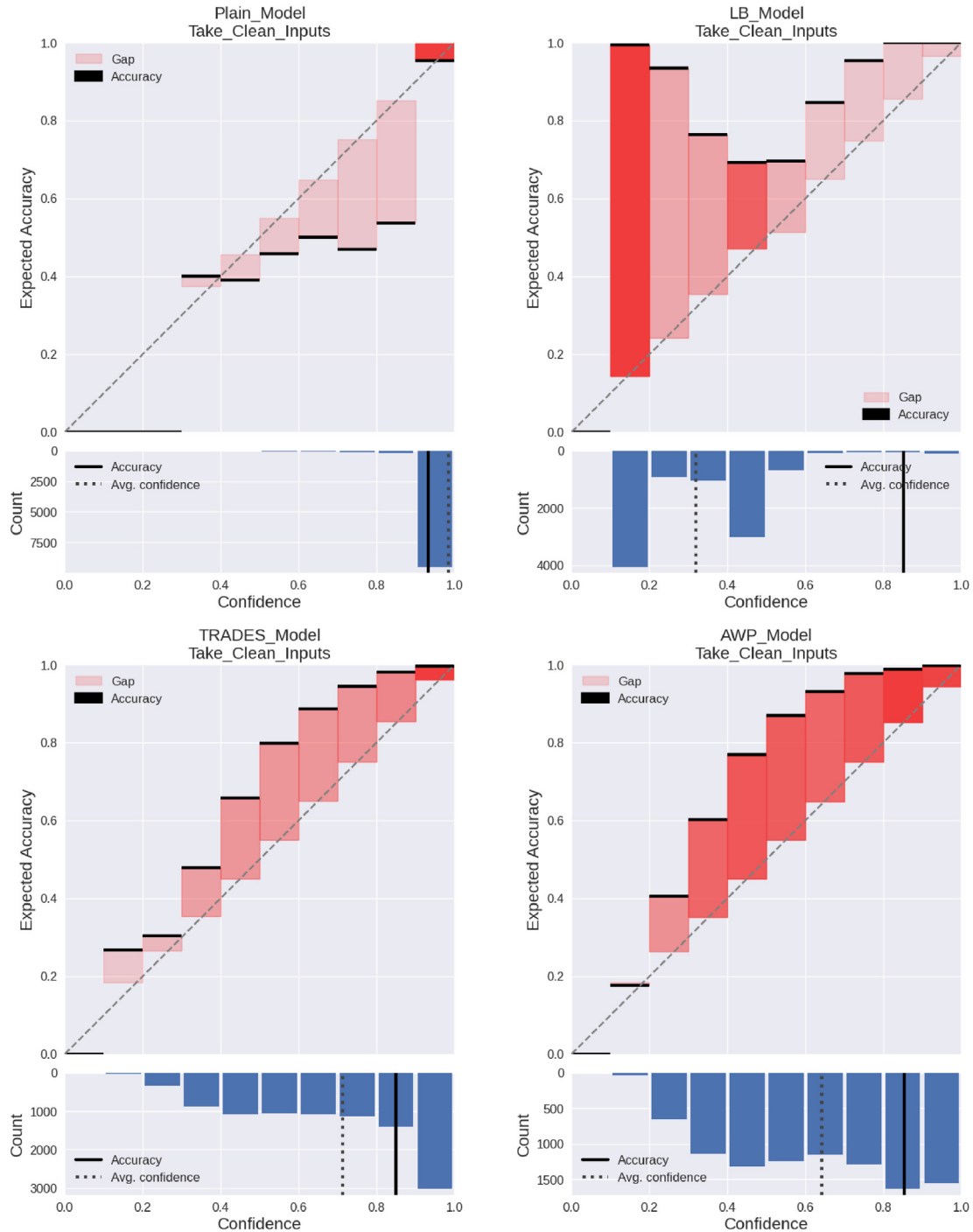


Fig. 3. Reliability diagrams and confidence histograms on CIFAR-10.

4.2. Robustness Diagrams

The reliability diagrams can indirectly illustrate the adversarial robustness of DNN classifiers. In order to provide a deeper analysis, interpretation and visualisation of the adversarial robustness of DNN classifiers, a novel diagram called robustness diagram is proposed based on the reliability diagram. The robustness diagrams plot the expected attack success rate of modified inputs as a function of confidence perturbation of adversarial examples.

Let $\{s_1, s_2, \dots, s_K\} \in S$ denote the Softmax confidence of a K-class classifier taking a clean input x , and $\{s'_1, s'_2, \dots, s'_K\} \in S'$ denote the Softmax confidence of the K-class classifier taking x' , the adversar-

ial example of x , as input. Then the confidence perturbation CP of x' against x is calculated according to:

$$CP = \sum_{n=1}^K |s'_n - s_n| \div 2 \quad (8)$$

such that $0 \leq CP \leq 1$.

Fig. 4 illustrates the robustness diagrams of three defence models and the plain model under the PGD attacks on the CIFAR-10 dataset with perturbation $\epsilon = 8/255$. The PGD attacks have a 100% success attack rate on the plain model, and nearly 8,000 out of 10,000 adversarial examples have over 90% confidence perturbation. In contrast, the confidence perturbations of the three

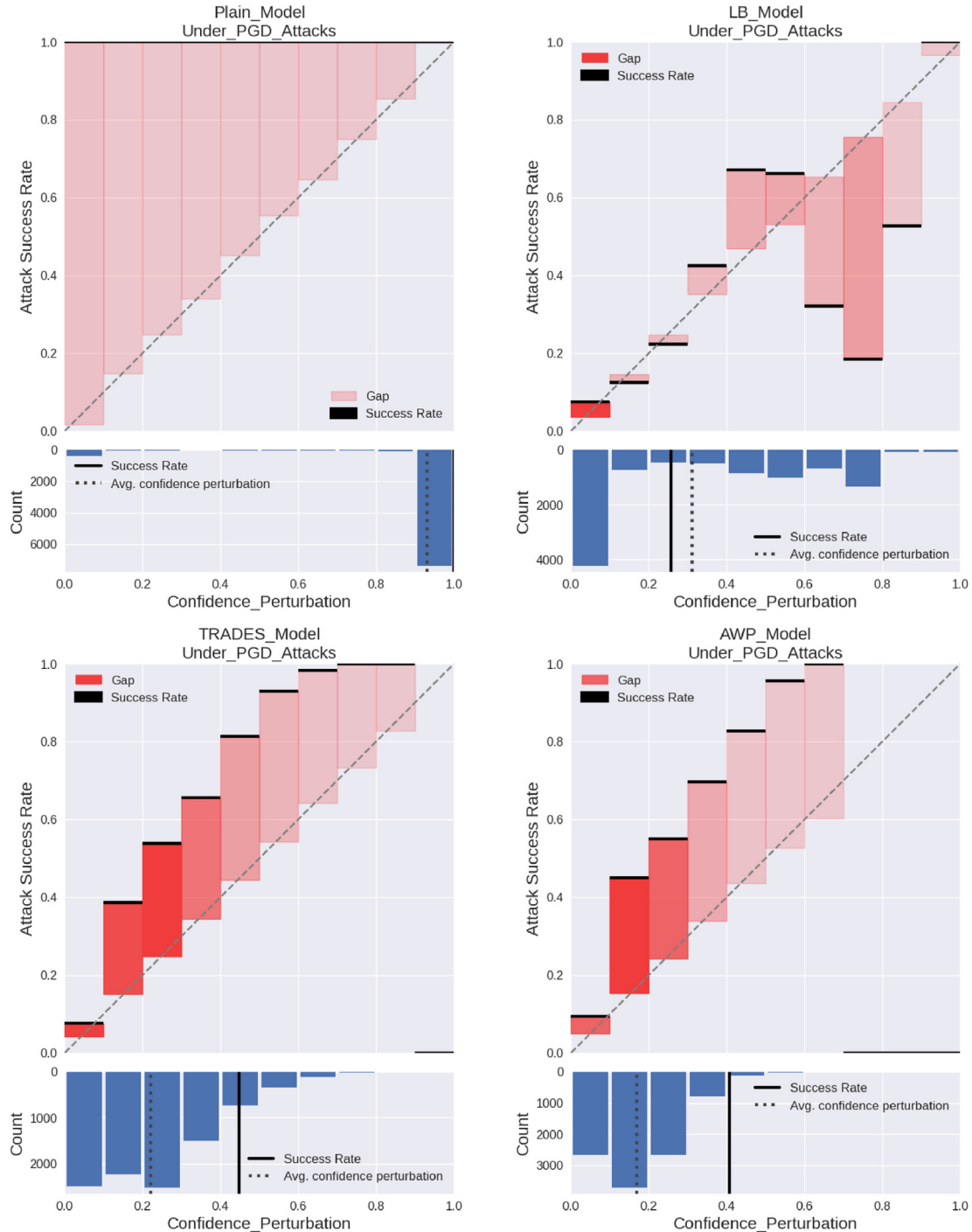


Fig. 4. Robustness diagrams and confidence perturbation histograms on CIFAR-10 under PGD attacks.

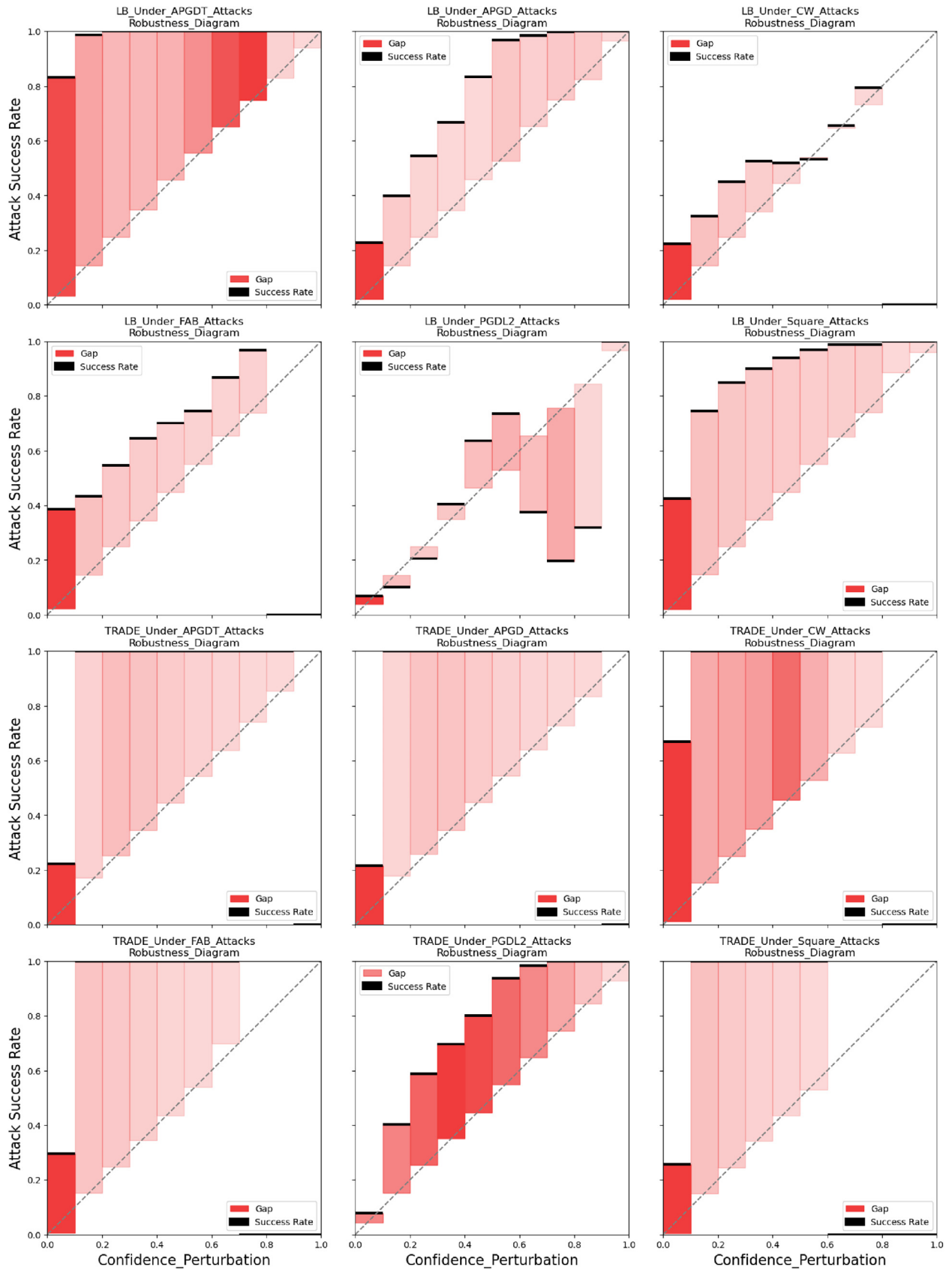


Fig. 5. Robustness diagrams of LB model and TRADES model under various adversarial attacks on CIFAR-10.

defence models are primarily below 40% when attacked by the PGD algorithm. It must be emphasised that only the LB model can have a lower average attack success rate than the average confidence perturbation. For example, the inputs modified by the PGD method that caused 70% to 80% confidence perturbation can only have nearly 20% attack success rate. This phenomenon proves that the method proposed in this paper does improve the adversarial robustness of DNN by balancing the confidence distribution of the incorrect classes.

Fig. 5 shows the robustness diagrams of the LB model and TRADES model under six different adversarial attack methods with perturbation $\epsilon = 8/255$. The colour depth of each bar represents the number of samples in the corresponding bin. According to the robustness diagrams, the robustness of the TRADES model is mainly from limiting the confidence perturbation of adversarial examples. In most instances, the robustness of the TRADES model is valid only when the confidence perturbation is less than 10%. Except for the PGD and PGD-L2 attacks, the other tested attacks had a 100% success rate on the TRADES model once they could cause a confidence perturbation of more than 10%. Furthermore, the robustness diagram provides a new perspective on the adversarial robustness of DNNs. The robustness of DNNs against adversarial examples can be enhanced by reducing the attack success rate of modified inputs with high confidence perturbation.

5. Proposed Adversarial Example Detection Method

Proactive defending methods such as adversarial training and regularisation can only provide limited adversarial robustness and can be defeated by adaptive attacking algorithms. The other intuitive direction of improving the robustness and trustworthiness of deep neural networks is developing attacks detectors running along with deep learning systems. The detectors can recognise whether the inputs are clean data or adversarial examples. Furthermore, the ideal detector should be able to determine the attack method that is used to generate the adversarial examples. This section aims to study the adversarial attack detection and detection robustness of the model trained by the proposed defence method in Section 3.

5.1. Design of Adversarial Example Detector

According to the knowledge described in the previous sections and the current adversarial example detection methods [31,32,34,35,50], it is intuitive to have an assumption that it is possible to train a sub-classifier to detect clean inputs and adversarial

examples that are created by various attack methods. Based on this assumption, a new Log-Softmax-pattern-based detection method is proposed to determine whether a batch of inputs is clean data or modified adversarial examples. If the inputs are adversarial examples, the detector can also estimate the perturbation algorithm that the attacker used.

As illustrated in Fig. 6, for the logit balancing model, the average values of the Min–Max Log-Softmax values of each mini-batch (batch size equals 100) of test data under seven adversarial attack methods are separately distributed. It can intuitively tell whether the logit balancing model is under attack and identify the attack methods by observing the abnormal Log-Softmax values. However, it is difficult to know the type of adversarial examples attacking the TRADES model by investigating their Log-Softmax values.

Without computing directionality toward the true label in [34] and the average logits of multiple noised inputs in [35], the proposed detection method takes the average Log-Softmax values of mini-batch data as inputs and recognises whether the mini-batch data are adversarial examples; if so, estimates the adversarial perturbation algorithm. The proposed detection method uses Log-Softmax values instead of logits because the self-adaptive logit balancing model was trained and regularised based on the mini-batch Log-Softmax values. Details of the proposed detection method are described in the following.

Let f_{LB} denote the pre-trained deep learning model that is regularised by using the method in Algorithm 1. Let $X = \{x_1, x_2, \dots, x_N\}$ denote a batch of inputs to f_{LB} , where x_i is a data instance of mini-batch X , and the N denotes the batch size. Let LS denote the output of the Log-Softmax function, such that:

$$LS = \text{LogSoftmax}(f_{LB}(X)) = \{ls_1, ls_2, \dots, ls_K\} \quad (9)$$

Collect the Log-Softmax values of every data instance within the mini-batch and rank the Log-Softmax values increasingly. Then compute the average Log-Softmax values channel by channel from the smallest to the largest. Such that there is a vector $\bar{ls}$ which has K real numbers, where K is the number of classes that f_{LB} needs to recognise.

We propose to train a multi-layer perceptron (MLP) - f_{MLP} as the sub-classifier with input–output pairs:

$$\{(\bar{ls}_0, 0), (\bar{ls}_1, 1), (\bar{ls}_2, 2), \dots, (\bar{ls}_A, A)\}$$

where $(\bar{ls}_0, 0)$ is the labelled Log-Softmax value pattern of clean inputs, and A is the number of different attack algorithms that the MLP can detect.

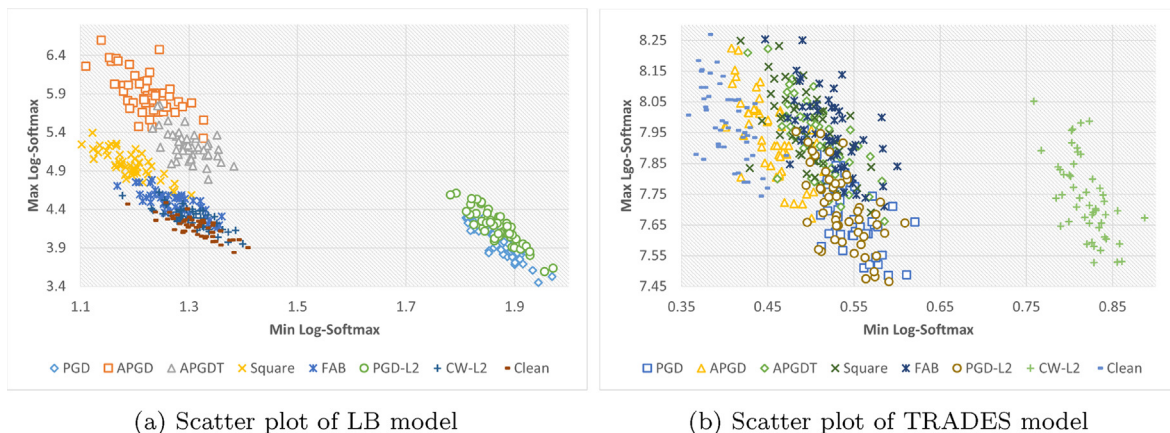


Fig. 6. The distribution of the Min–Max Log-Softmax values under various attacks [36].

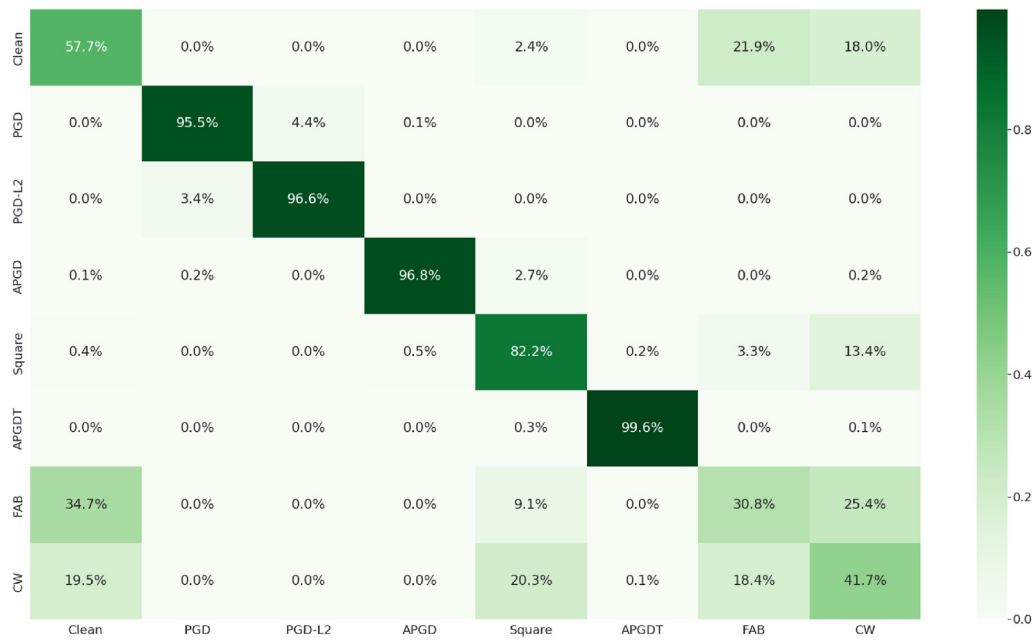
The reason for choosing MLP is that it can perform multi-classification tasks simply and efficiently. More importantly, the adversarial examples of this MLP can be found by using the gradient-based white-box attack methods on the MLP, which have the Log-Softmax value patterns similar to the patterns of clean inputs. These adversarial examples could simultaneously fool the deep learning model and adversarial example detector.

5.2. Logit Mimicry Attack for Proposed Detector

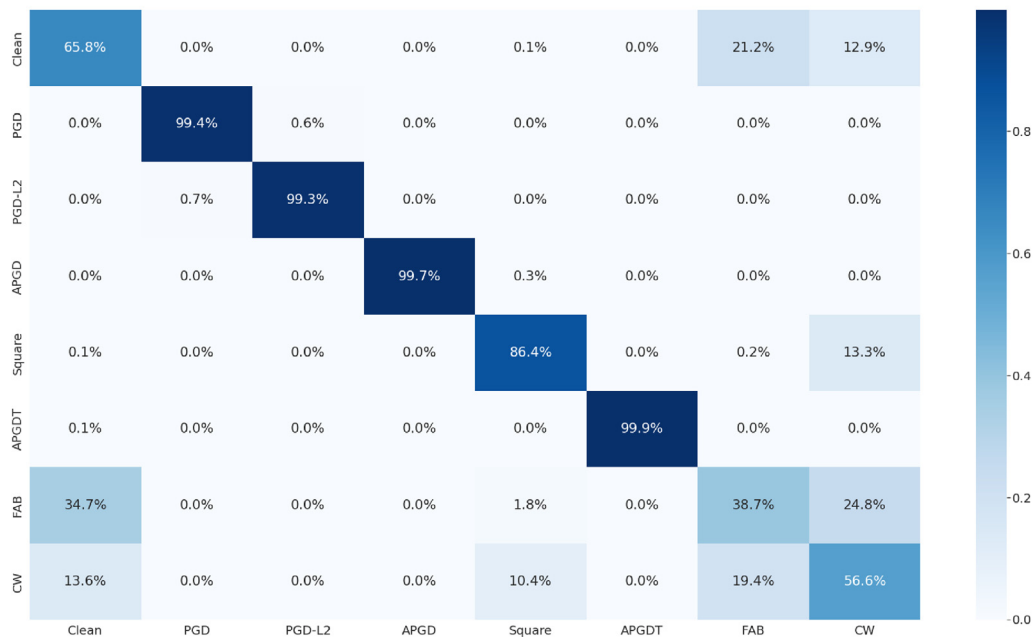
According to [35], both statistical tests and classifier-based detectors lack robustness under logit mimicry attacks (LMAs). However, the DNNs used to test their algorithms and methods

are plain networks without any defences, neither adversarial training nor regularisation. Therefore, their experimental conclusions are likely to be unilateral. Moreover, their detection methods are designed based on binary detectors, meaning their detectors can only output yes or no and cannot recognise the attack methods.

Because the proposed adversarial examples detection method is based on the classic MLP without any defences, this detector will be easy to fool by gradient-based white-box attacks, for instance, the FGSM. Therefore, it is easy to apply targeted attacks and find the adversarial perturbations P_D , such that: $f_{MLP}(\text{LogSoftmax}(f_{LB}(X + P_D))) = 0$, then force the proposed detector to believe all inputs are benign, where f_{MLP} is the MLP-based adversarial exam-



(a) Detection results of batch size equals to 60.



(b) Detection results of batch size equals to 120.

Fig. 7. Confusion matrices of the proposed adversarial example detection method at batch sizes equal 60 and 120.

ple detector, and LogSoftmax is the Log-Softmax output of f_{LB} regarding the input.

For evaluating the robustness of the proposed detection method under adversarial attacks, A LMA method is proposed to work with various adversarial example generation algorithms that try to mislead both the perturbation detector and the deep learning model at the same time. The LMA is described in the following:

Let f_{LB} denotes the pre-trained deep learning model that is regularised by using the Algorithm 1. Let f_{MLP} denotes the pre-trained MLP model that can recognise clean inputs and various adversarial examples. Let P_A denote the adversarial perturbations generated by any attack algorithm, where A is the label of the attack algorithm. Such that: $f_{LB}(X + P_A) \neq Y_{true}$, where Y_{true} is the ground-truth label of input X . Let P_D denotes the adversarial perturbations that are generated by any attack algorithm, such that $f_{MLP}(\text{LogSoftmax}(f_{LB}(X + P_D))) = 0$, where $\text{LogSoftmax}(f_{LB}(X + P_D))$ is the log-softmax output of f_{LB} regarding the input $X + P_D$. Then test the combined adversarial perturbations to check if they can have the following outcome:

$$\begin{cases} f_{LB}(X + P_A + P_D) \neq Y_{true}, \text{ AND} \\ f_{MLP}(\text{LogSoftmax}(f_{LB}(X + P_A + P_D))) = 0 \end{cases} \quad (10)$$

where $(P_A + P_D) < \epsilon$ and ϵ is defined in threat model to limit the amplitude of perturbations.

5.3. Experimental Result and Evaluation

5.3.1. Attack Detection via MLP Classifier

As illustrated in Fig. 7, the proposed detection method achieved very high accuracy on the CIFAR-10 data, above 95.5% setting batch size to 60 and over 99.3% setting the batch size to 120, when detecting gradient-based white-box attacks. The experiment results prove that the logit odds-based adversarial detection methods can perform binary classification to clean data and adversarial examples and effectively identify the generation methods of adversarial examples. The proposed detection method is outstanding for recognising gradient-based attacks. When using batch size 120, the proposed method can achieve over 99% accuracy with a 0% false-negative rate for detecting PGD, PDG-L2, APGD and APGDT.

FAB and CW are high-risk attack methods because the inputs modified by FAB and CW can be confused with clean images. With batch size 60, the detection accuracy of clean images is 57.7%, then

2.4% are misclassified as Square attacks, 21.9% are misclassified as FAB attacks, and 18.0% are misclassified as CW attacks. Meanwhile, 34.7% of FAB attacks and 19.5% of CW attacks are recognised as clean images. This will considerably threaten the proposed method because FAB and CW can avoid detection at a high enough rate. The confusion issues among clean images, FAB attacks and CW attacks cannot be improved by using a larger batch size. The false-negative rate of FAB attacks stays the same, and the false-negative rate of CW attacks drops from 19.5% to 13.6% by doubling the batch size to 120.

5.3.2. Comparison with Current Detection Methods

Table 7 shows the comparison of the current adversarial example classifiers and our proposed method. There are three advantages of the proposed method. First, using a multi-class detector instead of a binary classifier, the proposed method can recognise various adversarial attacks and their generation algorithms through one model. Second, the Log-Softmax-pattern-based detection is succinct. It does not require data reconstruction, noise injection or other complex computations. Third, the recognisable Log-Softmax patterns of different adversarial examples do not rely on adversarial training; therefore, they have better generality and scalability. Our proposed detection method achieved better performance detecting white-box gradient-based attacks compared with LNG [51], the state-of-the-art adversarial example detection method. The limitation of the proposed method is that it requires a small batch of data for adversarial attack detection, but other methods can do single-instance detection.

5.3.3. Detection Robustness Under Adversarial Attacks

After demonstrating the outstanding performance of the proposed adversarial example detection method of recognising multiple attacking algorithms, it is necessary to evaluate the robustness of this detection method. The LMA algorithm is applied to six different adversarial example generation methods and tries to create adversarial examples that can simultaneously mislead the detector and the deep learning model with self-adaptive logit balancing.

Table 8 shows the evaluation results by comparing the robustness accuracy of the deep learning model with self-adaptive logit balancing under six attacks and three levels of inputs. The first row of data is the robust accuracy of the deep learning model under different original attacks. The second row of data is the

Table 7

Performance comparison to the state-of-the-art adversarial example detectors [51].

Detection Method	Classifier	PGD	CW	Auto Attack	Square	FAB
Magnet [52]	Binary	-	89.02	-	-	-
Log Odds [34]	Binary	99.1	91.6	-	-	-
DkNN [53]	Binary	51.18	60.64	52.11	59.51	70.11
Knn [54]	Binary	54.46	65.25	52.64	73.39	59.75
LID [55]	Binary	67.95	55.60	56.26	85.93	99.48
Hu [56]	Binary	58.55	90.99	53.54	95.83	90.71
LNG [51]	Binary	91.39	89.74	84.03	98.82	99.98
Our Method (Batch size = 60)	Multi class	97.3	41.7	97.9	81.0	30.8
Our Method (Batch size = 120)	Multi class	99.4	56.5	99.9	87.5	38.7

Table 8

Accuracy comparison between original attacks and logit mimicry attacks.

	PGD	APGD	APGDT	SQUARE	FAB	PGD-L ₂	CW-L ₂
Original attacks	74.67	48.18	3.97	40.06	53.80	73.30	74.48
GN masked attacks	77.82	61.58	32.64	73.47	80.25	79.06	83.52
LMA	80.16	75.76	34.35	71.36	61.69	79.08	82.87
bypassing rate	100	98	1	93	96	3	100

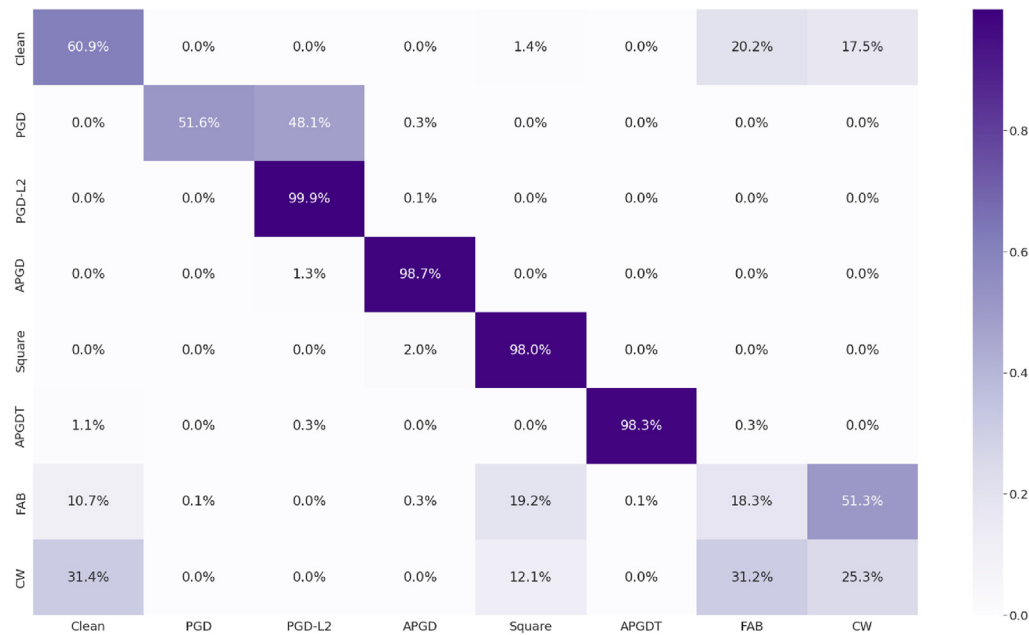


Fig. 8. Confusion matrix of the proposed method for classifying adversarial attacks with a perturbation limit $\epsilon = 16/255$ but use the perturbation limit $\epsilon = 8/255$ for training data.

robust accuracy of the deep learning model when using random Gaussian noise to cancel the effective perturbations. The third row of data is the robust accuracy of the deep learning model under adaptive logic mimicry attacks, and the fourth row is the bypassing ratio to the detector.

The LMA can mislead the detector to recognise PGD, APGD, SQUARE, FAB and CW attacks as clean images with a high false-negative rate but trade-off their adversarial effectiveness against the deep learning model. The robustness accuracy increased from 74.67% and 48.18% to 80.16% and 75.76% for PGD and APGD and increased from 40.6% to 71.36% and 53.90% to 61.69% for SQUARE and FAB. The robustness accuracy of CW attacks under the LMA setting is increased to 82.87%, which is close to the LB model's natural accuracy. On the contrary, the LMA can only achieve 1% and 3% false-negative rates for APGDT and PGD-L2 attacks, making the adversarial attacks less effective. The robust accuracy under APGDT attacks rose from 3.97% to 34.35%, and The robust accuracy under PGD-L2 attacks rose from 73.30% to 79.08%.

Unlike the experiment results in [35], the LMA can not always bypassing the proposed detection method and keep the adversarial perturbations to be effective at the same time. The experiments in this section show that forcing the adversarial examples to have similar Log-Softmax patterns of the clean inputs will remove the effectiveness of the adversarial perturbations. This phenomenon can help researchers further understand DNNs and design more robust algorithms.

In addition to the LMA, we tested the pre-trained adversarial example detector using the adversarial attacks generated with higher perturbation budget. As shown in Fig. 8, the detector's performance was not significantly confiscated. The false positive and the false negative rate were not increased compared with Fig. 7 (a). The only issue is that more significant perturbations can cause PGD attacks to be misclassified as PGD-L2 attacks.

6. Summary

This paper studies the issues of adversarial robustness of DNNs against human-designed adversarial perturbations. The proposed defence method focuses on improving the robustness of DNNs

against adversarial attacks via self-adaptive logit balancing and Gaussian noise boost training. The experimental results of this paper have been evaluated carefully and rigorously, and the results outperformed adversarial training defence approaches under non-targeted attacks. Meanwhile, the proposed reliability diagram can analyse and visualise the robustness of DNNs under adversarial attacks and give a new insight into how to improve adversarial robustness next. Moreover, the proposed detection approach can achieve high accuracy and precise adversarial attack recognition without complex computation. The proposed attack detection method is state-of-the-art in identifying white-box gradient-based attacks. Unlike the results described in [35], the logit mimicry attack can not easily fool the deep learning model and attack detector based on the self-adaptive logit balancing method. The trade-off between the false-positive rate of the detector and the fooling ratio of the deep learning model is inevitable with logit balancing. In the future, pre-trained DNNs with logit balancing can be used as a data augmentation tool, which can improve the label quality by replacing the one-hot label vectors with the Softmax vectors produced by the pre-trained logit balancing model.

CRediT authorship contribution statement

Jiefei Wei: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Resources, Writing – original draft, Writing – review & editing. **Luyan Yao:** Conceptualization, Methodology, Validation, Writing – review & editing. **Qinggang Meng:** Supervision, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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